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Expressions and Data Types

Introduction to Computer Programming

Management and Artificial Intelligence

Why Python?

- A **high-level** language for **beginners**
- with a **clean syntax**, but...
- **Industrial strength** programming language used at thousands of companies (one of three official Google languages)
- Free, well documented, well supported

Recall: We will use Python 3 (there are non-backwards compatible differences with respect to Python 2). Further, Python 2 is now unsupported.

Expressions and Statements

A (Python) program is composed by a sequence of:

- **definitions** (also **expressions**) which are **evaluated**
- **statements** (also **commands**) which are **executed**

In plain terms...

- **expressions** yield some results (e.g., `5 * 5`)
- **statements** instruct the interpreter to do something (e.g., print on screen, assign to variable)

Expressions and statements can either

- be directly written into a shell (**interactive mode**)
- be written into a text file and then loaded into the shell (**script mode**).

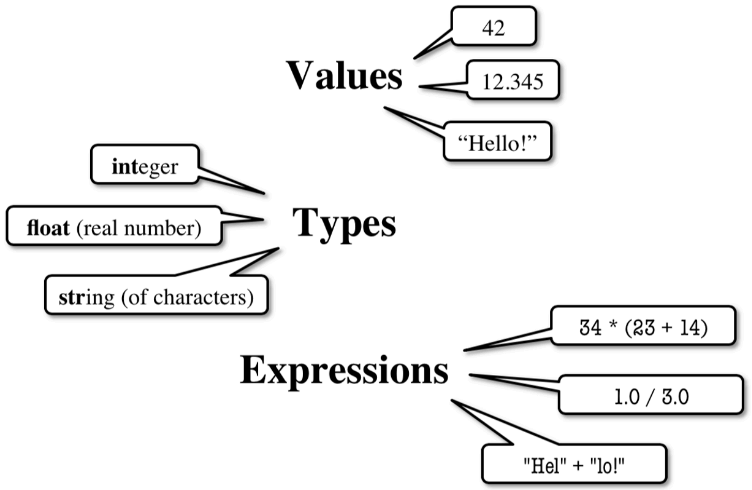
Interactive vs. Script Mode

- **Interactive Mode**: perfectly fine for testing and learning.
  Type your commands/expressions into the shell and Python will execute/evaluate those immediately:

`>>> 5 * 5`	←input
`25`	←output

- **Script Mode**: is what you normally do.
  - write the entire source code in a text file ending with the `.py` extension (say `myscript.py`)
  - `myscript.py` will be executed in its entirety by the Python interpreter

Programming in Python



- programs manipulate **data objects**
- each data object has its own **type**, which defines the type of operations that can be done with them
- objects can either be:
  - **Scalar**: cannot be subdivided
  - **Non-scalar**: have an inner structure that can be accessed

Scalar Objects

- `int`: represents integers (e.g., `5`)
- `float`: represents real numbers (e.g., `3.14`)
- `bool`: represents Boolean values (`True` or `False`)
- `NoneType`: special with only one value (`None`)

Hint: Use the `type()` function if you are not sure about the data type.

Examples:

>>> type(109)	←input
<class 'int'>	←output

>>> type(10.9)	←input
<class ’float’>	←output

Type `int()`

Values: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$

Integer literals look like a sequence of numbers with no periods nor commas.

Operations:

- `+`, `-`, `*`, `/`, `//` (floor division), `%` (modulus, remainder), `**` (exponentiation)
- `-` (unary minus, to negate a number)

Examples:

`>>> 7/5`	$\leftarrow$input
`1.4`	$\leftarrow$output
`>>> 7//5`	$\leftarrow$ `//` is the floor division!
`1`	$\leftarrow$output

Note In Python 3, operations between integers may not yield an integer.

Type `float()`

Values: some (approximated) range of $\mathbb{R}$
Note that:

- numbers with a "." are decimals (e.g., `2.0`)
- numbers with no decimal separator are integers (e.g., `2`)

Operations:

- `+`, `-`, `*`, `/`, `%` (modulus, remainder of division)
- `**` (exponentiation)
- `-` (unary, invert sign)

Operations between floats or between a float and an integer, always yield a float.

`>>> 7.0/5.0`	$\leftarrow$input
`1.4`	$\leftarrow$output
`>>> 7.0//5`	$\leftarrow$ `//` is the floor division!
`1.0`	

Python stores floating point numbers as base 2 (binary) fractions:

$$\{\text{number}\} = \{\text{integer mantissa}\} \cdot 2^{-\{\text{exponent}\}}$$

For example: $0.125 = 1 \cdot 2^{-3} = 1 \cdot \frac{1}{2^3} = 1 \cdot \frac{1}{8}$.

Warning: Not all numbers can be represented in base 2 fractions. Python chooses the best approximation it can. The same also holds for some numbers in base 10 fraction: e.g., what about 1/3?

Python may yield some **approximation errors**, that propagate as expressions go through.

Example:

`>>> 0.1+0.2`	$\leftarrow$input
`0.30000000000000004`	$\leftarrow$output
`>>> 0.1+0.2+0.1+0.2`	$\leftarrow$input
`0.6000000000000001`	$\leftarrow$output
`>>> 0.1+0.2+0.1+0.2+0.1+0.2+0.1+0.2`	$\leftarrow$input
`1.2000000000000002`	$\leftarrow$output

Type `bool`

Values:

> `True` and `False` (capitalized!)

Operations:

$$\texttt{not a} = \begin{cases} \texttt{True} & \text{if } \texttt{a} \text{ is } \texttt{False} \\ \texttt{False} & \text{if } \texttt{a} \text{ is } \texttt{True} \end{cases}$$

$$\texttt{a and b} = \begin{cases} \texttt{True} & \text{if both } \texttt{b} \text{ and } \texttt{a} \text{ are } \texttt{True} \\ \texttt{False} & \text{otherwise} \end{cases}$$

$$\texttt{a or b} = \begin{cases} \texttt{True} & \text{if } \texttt{a} \text{ is } \texttt{True} \text{ or } \texttt{b} \text{ is } \texttt{True} \\ \texttt{False} & \text{otherwise} \end{cases}$$

Booleans are the output of comparisons between items (e.g., `int` or `float`):

- Order: `i<j`, `i>j`, `i<=j`, `i>=j`
- (In)equality: `i!=j`, `i==j`

Truth Table between two Boolean variables

`a`	`b`	**not** `a`	**not** `b`	`a` **and** `b`	`a` **or** `b`
`True`	`True`	`False`	`False`	`True`	`True`
`False`	`False`	`True`	`True`	`False`	`False`
`True`	`False`	`False`	`True`	`False`	`True`
`False`	`True`	`True`	`False`	`False`	`True`

Type Conversions

Switching between `float` and `int` can be performed via **casting** (**explicit** conversion):

- `float(2)` converts value `2` (integer) to `2.0` (floating point)
- `int(2.9)` converts value `2.9` (floating point) to `2` (integer)

> **Warning** `int()` does not round to the nearest integer: rather, it truncates the decimal part. To round to the nearest integer, use `round()` instead.

Python supports **widening** automatically (**implicit** conversion):

$$\underbrace{\texttt{bool} \rightarrow \texttt{int} \rightarrow \texttt{float}}_{\text{narrow to wide}}$$

- **Widening**: Python does it automatically, if needed. Example:

  `1/0.5=2.0` because Python casts `1` to `float`

- **Narrowing**: Python never does it automatically. Because narrowing leads to information loss!

Expressions

- **Expressions** are combinations of **objects** and **operators**
- an expression yields a **value**, which has its own **type**
- the simplest expression has the form:

```
<object, operator, object>
```

Arithmetic Operations on `int` and `float`: a Recap

Syntax	Operator	Output Type
`i+j`	Sum	if both of them are floats, result is float
`i-j`	Difference	if both of them are int, result is int
`i*j`	Product	if either of them is float, result is float
`i/j`	Division	result is float (always)
`i%j`	Remainder (Modulo)	if both of them are floats, result is float
`i**j`	Power of	if both of them are int, result is int
`i//j`	Floor division	if either of them is float, result is float

Comparison Between `int`s and `float`s: a Recap

NOTE: Comparisons below evaluate to a `bool`.

Expression		Evaluation
`i<j`	$\rightarrow$	`True` if `i` is smaller than `j`
`i<=j`	$\rightarrow$	`True` if `i` is smaller than or equal to `j`
`i>j`	$\rightarrow$	`True` if `i` is greater than `j`
`i>=j`	$\rightarrow$	`True` if `i` is greater than or equal to `j`
`i==j`	$\rightarrow$	`True` if `i` is equal to `j`
`i!=j`	$\rightarrow$	`True` if `i` is not equal to `j`

Chained Comparisons

Comparisons can be chained arbitrarily and comparisons have the same priority:

`i<j<k` is the same as `i<j and j<k`

Since comparisons do not act in a pairwise fashion, *weird* comparisons like

`i<j>k`

are perfectly valid since it is equivalent to `(i<j) and (j>k)`.

Operator Precedence

Precedence between operators can be *forced* via parenthesis.

In principle, operators follow the **PEMDAS** order:

1. **P**arenthesis
2. **E**xponential
3. **M**ultiplication and **D**ivision
4. **A**ddition and **S**ubtraction

The Big Picture of Operator Precedence

1. Parenthesis: `(...)`
2. Exponentiation: `**`
3. Unary operators: `+` and `-`
4. Binary arithmetic: `*`, `/`, `//` and `%`
5. Binary arithmetic: `+` and `-`
6. Comparisons: `>`, `<`, `>=`, `<=`, `==`, `!=`
7. Logical `not`
8. Logical `and`
9. Logical `or`

NOTE: Same line means same precedence. Read ties left to right.

HINT: In case of emergency, use parenthesis!